Object-Oriented System Analysis Design

11th November 2024

Marking Scheme

This marking scheme has been prepared as a **guide only** to markers. This is not a set of model answers, or the exclusive answers to the questions, and there will frequently be alternative responses which will provide a valid answer. Markers are advised that, unless a question specifies that an answer be provided in a particular form, then an answer that is correct (factually or in practical terms) **must** be given the available marks.

If there is doubt as to the correctness of an answer, the relevant NCC Education materials should be the first authority.

Throughout the marking, please credit any valid alternative point.

Answer ALL questions

Marks

Question 1

a) Define the following terms in the context of Object-Oriented Analysis and Design:

i) Inheritance **2**

Mark scheme
Inheritance allows subclasses to possess traits of a superclass such that they can reuse existing superclass data and procedures. (1 mark) Also provides mechanism for propagating changes. (1 mark)

ii) Product backlog grooming in the context of Agile development. **2**

Mark scheme
Products go to the top of the backlog when they are in a 'ready' state which means that they are sufficiently defined and well understood. Some teams set a criterion for this. (1 mark)
Grooming refers to activity of creating and refining product backlog items, estimating them, refining and prioritising them. (1 mark)

iii) MoSCoW based requirements specification. **2**

Mark scheme
MoSCoW is a requirements prioritisation method (1 mark) that identifies requirements that a system Must-have, Should-have, Could-have, and Won't-have. (1 mark)

b) Multiple choice questions: Please select ONE (1) from the options provided.

i) When using Use-Case Points for estimation, which of the following can be used to assess technical complexity: **1**

A. Response time or throughput performance objectives
B. Lead analyst capability
C. Difficulty of programming language
D. All the above
E. None of the above

Mark scheme
A. response time or throughput performance objectives

ii) In object-oriented design, which of the following is a behavioural diagram? **1**

A. Class diagram
B. State diagram
C. Deployment diagram
D. Package diagram

Mark scheme
B. State diagram

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Marks

iii) In OO, a property of a class can be described by a **1**

A. Message
B. Instantiation
C. Method
D. Attribute
E. All of the above

Mark scheme
D. attribute

iv) In OO, an accessor method **1**

A. Changes the state of an object's attributes
B. Returns a result without affecting the state of the object
C. Is the same as a mutator method
D. Is the same as a command

Mark scheme
B. returns a result without affecting the state of the object

Total 10 Marks

Question 2

Outline FIVE (5) merits **and** FIVE (5) demerits of the Waterfall model. **10**

Mark scheme
Any 5 from each of the lists below or alternative valid answers.

Merits of Waterfall model

- ***Structured, linear approach means that the deliverables are more predictable and therefore easier to estimate and plan.***
- ***Defines end goal early. What you plan is what you get.***
- ***Provides clear structure and steps that help coordinating large projects and clarifying costs and schedules at the outset. There are no financial surprises at roll out stage.***
- ***Suitable for well-defined user requirements that must be defined at the outset and maintained till the end of the project.***
- ***Easier to assess fulfilment of requirements.***
- ***Insists on documentation to maintain development audit trail so development team can account for their activities.***

Demerits of Waterfall model

- ***Changes cannot be made once the requirements are agreed.***
- ***Needs requirements to be stated at the outset. Difficulty of accommodating change after the process is underway. Changing user needs can sometimes result in unexpected costs at the review stage. Requirements may not be clear, understood or available at the outset. Users do not always fully know what they need or what they can have until much later in the discussion or when they start using the system.***
- ***Technically, a phase has to be complete before moving onto the next.***

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Marks

- ***Late testing means that defects may not be discovered until it is too late but can be enhanced by a continuous testing approach as recommended in models like the V-Model or the W-Model.***
- ***Limited client involvement can result in no user buy in or inability to test understanding.***
- ***Systems are often out of date even as they are being designed.***

Award 1 mark for each point, up to 10 marks. 5 marks for merits and 5 marks for demerits.

Total 10 Marks

Question 3

a) List the FOUR (4) core values of the Agile Manifesto **and** explain how the Unified process (UP) satisfies the principles of the Agile Manifesto. **8**

Mark scheme

Agile Manifesto	***How Unified Process satisfies the Manifesto***
Individuals and interactions over processes and tools (1 mark)	***Whilst placing individuals and interactions at the forefront, UP insist on sequential steps broken into distinct phases which must each be completed before the next. (1 mark)***
Working software over comprehensive documentation (1 mark)	***UP insists on producing enough (not over detailed) documentation to ensure that stakeholders have up-to-date clear understanding of a project status, risks and requirements. Focus is on creating quality software through a disciplined approach that is iterative and incremental. (1 mark)***
Customer collaboration over contract negotiation (1 mark)	***Observes contractual obligations whilst placing customer communication and feedback at the forefront of development. A project manager creates a project plan and is responsible for ensuring delivery. (1 mark)***
Responding to change over following a plan (1 mark)	***Follows a plan but uses customer feedback to inform each increment. The change requests must be approved by the project manager. (1 mark)***

1 mark for each value and 1 mark for how the UP satisfied the principle of the Agile Manifesto.

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Marks

b) Name TWO (2) of the TWELVE (12) principles of the Agile Manifesto. **2**

Mark Scheme

- ***Customer Satisfaction***
- ***Welcome Change***
- ***Frequent Delivery***
- ***Collaboration***
- ***Motivated Individuals***
- ***Face-to-Face Conversation:***
- ***Working Software***
- ***Sustainable Development***
- ***Technical Excellence***
- ***Simplicity***
- ***Self-Organising Teams***
- ***Reflection and Adjustment***

1 mark for any correct principle named up to max of 2 marks

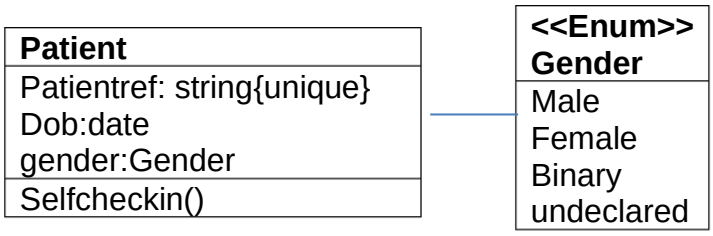
Total 10 Marks

Question 4

a) In a Health Care System, where patients can make several appointments over time, the gender of a patient can be recorded as *male*, *female, binary* or *undeclared*. **4**

Produce an implementation level class diagram showing the class Patient with at least THREE (3) attributes of different data type, ONE (1) of which must be gender as an enumeration. Show that the patient can *self-check-in* for an appointment.

Mark Scheme



Award 1 mark for the patient class and attribute names; 1 mark for the data types; 1 mark for the enumeration; 1 mark for the selfcheckin() method.

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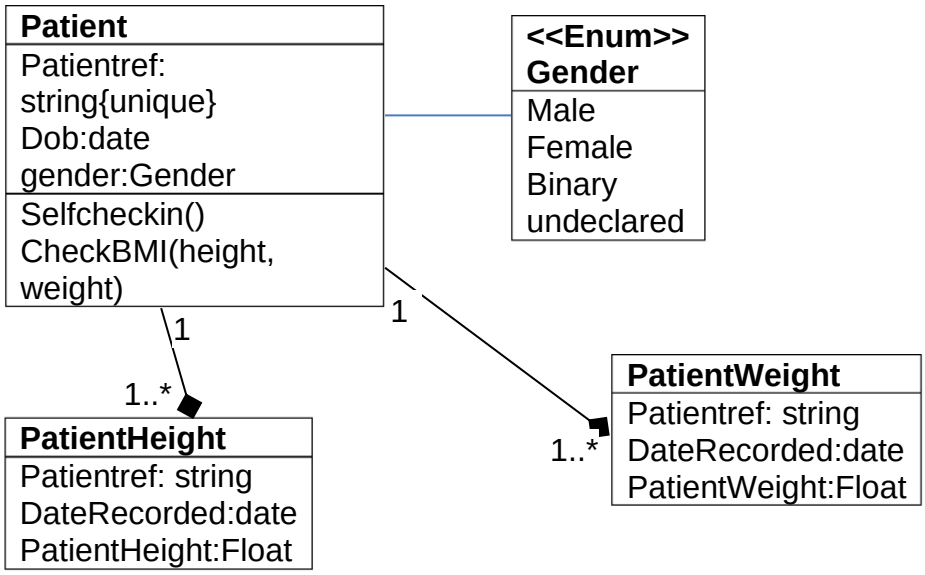
Marks

b) In the class diagram in Question 4a, above, explain with an example why patient height and patient weight are best placed in a separate class. **6**

Add the height and weight classes to the class diagram including the relationships to the Patient class. Show that height and weight can be used to check a patient's Body Mass Index (BMI).

Mark scheme

Height and weight are not shown in the Patient class because they are transient (1 mark). For example, they change over time unlike someone's date of birth which is permanent (1 mark).



Award up to 1 mark for explanation, 1 mark for the example, 1 mark for each of the classes, 1 mark for showing the relationships, 1 mark for the checkBMI() method

Total 10 Marks

Question 5

Identify **and** explain FIVE (5) reasons why a software development project might fail. **10**

Mark Scheme

Any 5 from below or from a host of alternative valid responses.

- ***Poor project management (response can include any of the following)***
  - ***Over budget***
  - ***Exceeds schedule and misses marketing window***
  - ***Unrealistic or unarticulated project goals***
  - ***Poor reporting of status***
  - ***Unmanaged risks***
  - ***Poor stakeholder communication***
  - ***Inability to handle complexity***
  - ***Poor methods, methodologies, process***

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Marks

- ***Wrong or inefficient set of development tools***

- ***Inadequate testing and documentation (too much or too little or irrelevant)***
  - ***Impracticality of testing all use cases, all data, all pathways***
  - ***Difficulty defining absolute correctness especially for non-functional requirements***
- ***Badly defined requirements***
  - ***Incomplete, erroneous, incomprehensible***
  - ***Does not comply with standards, practices, conventions, legislation***
- ***Does not meet customer / user requirements***
  - ***Low quality***
  - ***Performance not as expected***
  - ***Too difficult to use***
- ***Poor people management***
  - ***eReadiness: eLiteracy, time, management protocols***
  - ***eAwareness***
  - ***eGovernance***
  - ***proper people management involves understanding the strengths and weaknesses of team members and assigning tasks accordingly. Poor people management can lead to a mismatch of skills and tasks, resulting in inefficiencies and subpar work***

Award 1 mark for identification of the reason and 1 mark for explanation.

Total 10 Marks

Question 6

a) Describe THREE (3) types of object-oriented based partitioning used for testing. **6**

Mark Scheme

- ***State-based partitioning: operations that cause state changes are tested separately from those that do not.***
- ***Attribute-based partitioning: for each class attribute, operations are classified according to those that use the attribute, those that modify the attribute and those that do not use or modify the attribute.***
- ***Category-based partitioning: operations are categorised according to the function they perform: initialisation, computation, query, termination.***

Award 1 mark for naming the technique and 1 mark for the description (up to 6 marks)

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Marks

b) Describe TWO (2) types of object-oriented integration testing. **4**

Mark scheme

- ***Thread-based testing: tests all classes required to respond to one input or event.***
- ***Use-based testing: begins by testing independent classes and the dependent classes that make use of them.***
- ***Cluster testing: groups of collaborating classes are tested for interaction errors***

Award 1 mark for naming the technique and 1 mark for the description (up to 4 marks)

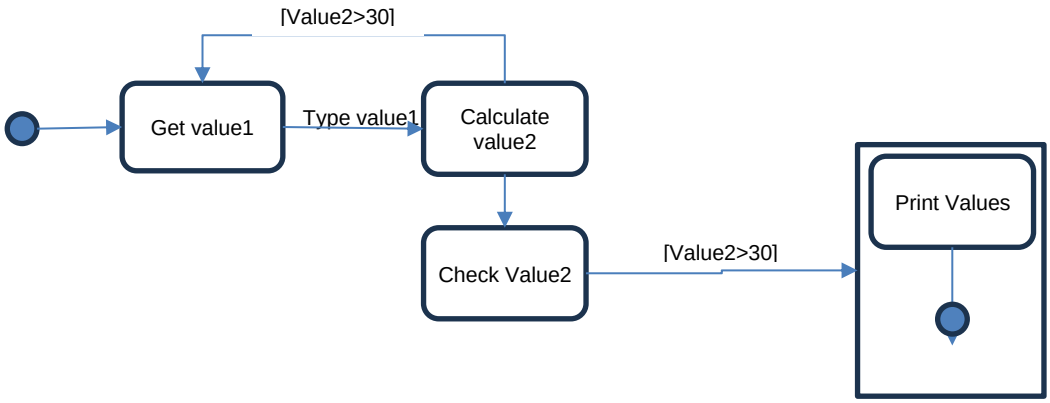
Total 10 Marks

Question 7

a) Draw a state diagram for the following piece of code logic: **7**

```
Get value1
Multiply value1 by 6 giving value2
   If value2 > 30 Print value1, value2
         Close
   Else
         Return to start
```

Mark scheme



States: Award 4 marks for all states (deduct 1 mark from the score for states if start and close nodes are not shown; use of decision diamond is acceptable).

Transitions: Award 1 mark for Value1 capture; Award 2 marks for condition.

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Marks

b) Provide THREE (3) of the purposes of UML sequence diagrams. **3**

Mark scheme

Any of the following (1 mark for each point)

- ***Document System Behaviour to help developers and stakeholders understand how different components in a system interact over time.***
- ***Provide a common language for developers, analysts, and stakeholders to discuss system behaviour and design, reducing misunderstandings.***
- ***Show the sequence of messages exchanged to clarify flow of control.***
- ***Break down complex Use Cases into simpler interactions.***
- ***Help to identify system components and their responsibilities.***

Total 10 Marks

Question 8

a) Identify FIVE (5) metrics for measuring inheritance. **5**

Mark scheme

Any 5 from the following:

- ***Number of root classes (NOR)***
- ***Fan in (FIN): maximum number of input signals***
- ***Number of children (NOC)***
- ***Depth of inheritance tree (DIT)***
- ***Method inheritance factor (MIF)***
- ***Number of classes with children/specialisations***

Award 1 mark for each metric (up to 5 marks)

b) Briefly explain what is meant by polymorphism and state ONE (1) advantage of polymorphism. **2**

Mark Scheme

Polymorphism allows different objects to respond to the same message in different ways i.e. applies functions/methods allowing a task to be performed in multiple forms/ways (1 mark).

Code Reusability: Polymorphism promotes code reusability by allowing the same code to work with different types of objects (1 mark).

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Marks

c) Provide ONE (1) polymorphism example based on a cooking app. **3**

Mark Scheme

Say, one wishes to create a program that will allow cooking of any dish. Every dish requires specific ingredients and utensils and will require different settings of the cooker.

(1 mark)

A superclass, Food, can be created which has a method cookdish(dishname) that takes on parameters (ingredients, utensils, setting…) from any dish to be cooked. ***(1 mark)***

The dishes are set up as subclasses that hold information about the required ingredients, utensils, setting e.g. rice class, porridge class, pasta class and so on. Each dish would have a cookdish() method which uses the method in the superclass so rice.cookdish() would get the ingredients, utensils and setting for cooking rice and pasta.cookdish() would get them for cooking pasta. ***(1 mark)***

Total 10 Marks

Question 9

a) Explain what is meant by a User Story in Agile. **4**

Mark Scheme

User stories are short informal, general explanations of a feature from the perspective of a user. Helps unpackage use cases by articulating how a piece of work will deliver value to the user. The stories can also be used to aid estimation. (1 mark) Captures single feature, use case, enhancement, role or scenario as follows: As a Role (Who), I want to action (what), so that benefits (why) (1 mark)

- ***Stories must be integrated into current Agile workflow. SCRUM adds the user stories to Sprints and burns them down over the duration of the Spring. Kanban adds user stories to the backlog and process them through the workflow.*** ***(1 mark)***
- ***During a sprint planning meeting, Agile team defines what user stories they will take to the Sprint. The Stories are scored on importance, complexity and completion time*** ***(1 mark)***

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Marks

b) Provide a User Story for a Warehouse Clerk who is responsible for fulfilling orders in an Order Processing System. **6**

Mark Scheme

Description of user role	***As a Warehouse Clerk***	***1 mark***
Users view of the system action	***I want to use the Order Processing System to fulfil orders, organise and replenish stocks.***	***1 mark***
Benefits to the User	***So that I can ensure availability of stocks to satisfy customer orders.***	***1 mark***
Perceived challenges	***Concurrent orders of the same product may sometimes create difficulty as one order could result in a change in stock levels that would impact on the other. (1 mark)*** ***Sometimes orders include a number of different products that may have different shipping needs. (1 mark)***	***2 marks***
Other users that use feature	***The system is also used by customers and orders clerks to submit orders and by management to check progress and produce monitoring reports.***	***1 mark***

Total 10 Marks

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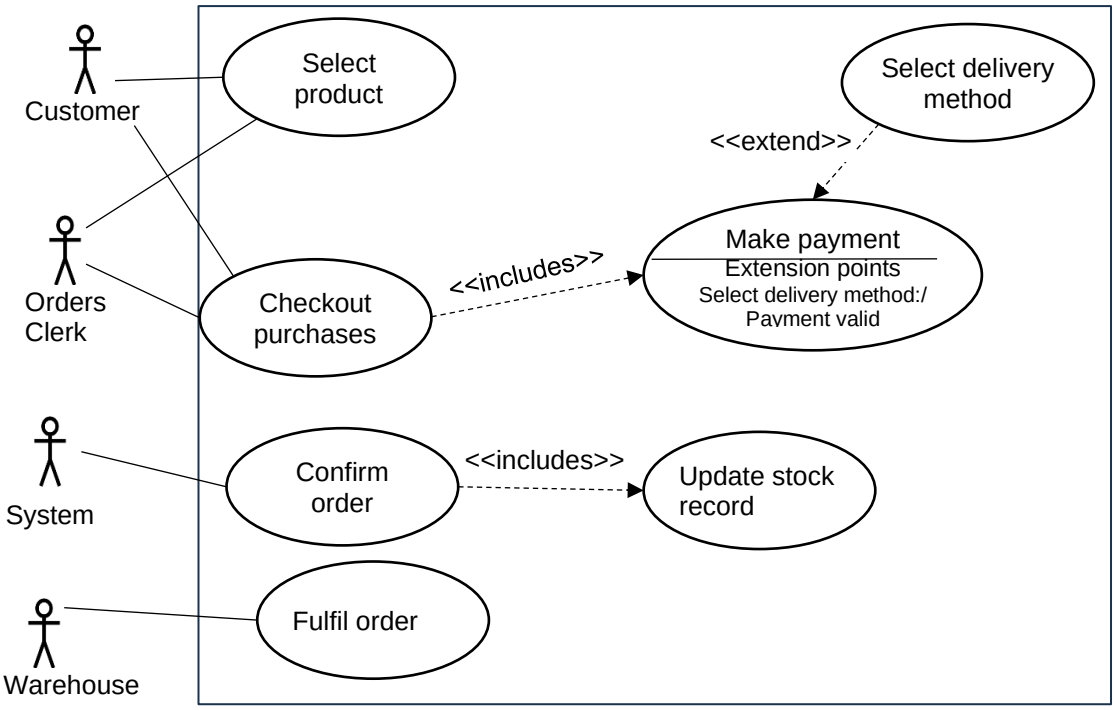
Marks

Question 10

Draw a UML Use-Case Diagram to illustrate the following process for an online purchase and delivery of products. Customers can make orders themselves or by phone. **10**

1. Customers or Orders Clerks select ONE (1) or more products from a product selection page.
2. Customers or Orders Clerks select checkout and enter payment details.
    a. If payment is not valid, customers are returned to the Checkout page
    b. If payment is valid, customers select a delivery method (collection or shipping) and complete appropriate details.
3. The system confirms the order and updates the stock record.
4. The warehouse fulfils the order.

Mark scheme



Award 3 marks for Customer and Orders Clerk actor and use cases; 2 marks for System actor and use case; 1 mark for warehouse actor and use case

Award 2 marks for <<extend>> and extension point

Award 2 marks for <includes>>

Total 10 Marks

End of paper

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Learning Outcomes matrix

Task	Learning Outcomes assessed	Marker can differentiate between varying levels of achievement
1	1-4	Yes
2	1	Yes
3	1	Yes
4	2, 3	Yes
5	1, 4	Yes
6	3, 4	Yes
7	2, 3	Yes
8	3, 4	Yes
9	2	Yes
10	2,3	Yes

Grade descriptors

Learning Outcome	Fail	Referral	Pass	Merit	Distinction
1. Discuss various approaches to systems analysis and design and explain their strengths and weaknesses.	Unable to identify and assess approaches.	In an extremely limited way, can identify and assess methodologies and methods to reach limited solutions.	Can adequately identify and assess appropriate methodologies and methods to reach appropriate solutions.	Can sufficiently identify and assess appropriate methodologies and methods to reach good solutions.	Can coherently identify and assess coherent methodologies and methods to reach well-defined and structured solutions.
2. Apply object-oriented techniques for systems analysis and modelling of well-defined but complex problems.	Unable to explain key principles or discuss developments.	In an extremely limited way, can explain principles and discuss developments.	Can adequately explain principles and discuss developments.	Has very good grasp of explain principles and discuss developments.	Can coherently discuss developments and explain and adapt principles.
3. Explain and use UML to represent object-oriented models.	Little to no understanding of UML or object-oriented models.	Limited understanding of UML or object-oriented models.	Adequate ability to apply UML with sufficient representations.	Demonstrates very good ability to apply and adapt UML models with a substantial system understanding.	Outstanding ability to apply UML and adapt models in a manner that demonstrates understanding of effectiveness of methods, actions and results.
4. Understand the integration between the design models and implementation.	Lacks awareness of different perspectives or approaches within the area of study.	Has extremely limited awareness of different perspectives or approaches within the area of study.	Has adequate awareness of different perspectives or approaches within the area of study. Understands requirements traceability but with some gaps.	Has very good, informed awareness of different perspectives or approaches within the area of study. Understands requirements traceability.	Excellent awareness of different perspectives or approaches within the area of study.

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